
From Pretrained Knowledge to Action: Auditing Language Models in a Deterministic Game

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Abstract

1 Pretrained language models can describe strategy without reliably converting that
2 knowledge into actions. Measuring this gap is difficult because evaluations may
3 confound the target capability with task semantics, interface, model authority,
4 response contract, and score. We present SLAY-BENCH, a deterministic Slay-
5 the-Spire-inspired simulator with four separately scored decision operations, two
6 characters, two state encodings, and seven open-model configurations. Differences
7 are operation-specific. Qwen3-235B-A22B improves fixed-deck card choice over
8 Qwen3-32B in all four character-encoding cells and 19/20 seed pairs, but not
9 uniformly in immediate sequencing, combat execution, or hybrid Act-1 rollout.
10 Crucially, an adversarial audit finds that always answering “Strike” solves all 40
11 removal fixtures and a deterministic card-name dictionary solves all 120 rotated
12 fixed-deck cases, without simulating a transition. These tests do not show that
13 the evaluated models used either shortcut; they show that the instruments do not
14 require planning. We quarantine removal, bound fixed-deck scores as recognition
15 and category-aligned choice, and retire the cross-task scalar and horizon interpreta-
16 tion. From this completed falsification we derive a same-state controlled-horizon
17 protocol and a reusable audit ladder: before benchmark scores are attributed to
18 planning, the instrument should first survive compute-free policies.

19 1 The measurement problem

20 Pretraining may encode game rules, card interactions, and strategic abstractions, yet an acting agent
21 must select a legal action under a concrete objective and carry its consequences forward. PTA
22 therefore needs measurements that distinguish *stored knowledge*, *action selection*, and *closed-loop*
23 *control*. Existing planning benchmarks offer exact symbolic evaluation [10], while game benchmarks
24 offer interactive breadth [3, 8]. But a line drawn across heterogeneous tasks can look like a “planning
25 horizon” even when task semantics, output contracts, baselines, and agent authority all changed.

26 We use a deterministic deck-building game to ask a narrower question: *where does pretrained*
27 *knowledge become useful for a specified decision operation?* Slay the Spire is not new as an
28 LLM testbed. MiniSTS studied language-driven play [1]; subsequent work examines card-rule
29 synergy [2] and original-game agents across broader suites [9]. Our contribution is an inspectable,
30 within-simulator *operation profile* and the adversarial audit that changed its interpretation.

31 **Contributions.** First, SLAY-BENCH makes four decision operations inspectable in native units and
32 exposes model authority. Second, a seven-configuration profile and matched Qwen3 comparison
33 show operation-specific differences, including a fixed-deck card-choice gain in 4/4 cells and 19/20
34 seed pairs without uniform transfer. Third and centrally, a completed adversarial audit supplies
35 two compute-free counterexamples, retracts unsupported scalar and horizon claims, and motivates a
36 controlled protocol and audit ladder for agent benchmarks.

Table 1: The four operations are not levels on one horizon. “Ref.” is the scoring reference; scripted choices are outside model control. Removal-v1 is quarantined.

Operation	Model-controlled output	Score / ref.	Supported interpretation
Immediate sequence Combat execution	ordered card indices for one state card plays each turn	damage / exhaustive best win; remaining HP / greedy bot	immediate-damage sequence quality closed-loop combat execution
Fixed-deck choice Hybrid Act-1 rollout	deck label; one of three cards combat and reward choices	authored label and pick floors and progress / matched greedy	taxonomy and card-category alignment behavior under scripted route, rest, shop, and events

2 A deterministic operation profile

SLAY-BENCH implements a deterministic, Slay-inspired simulator with independent random streams, two characters (Ironclad and Silent), Acts 1–3 mechanics, and two semantically matched prompt encodings: structured JSON and raw English. It is not claimed to be a bit-exact clone of the commercial game. Table 1 states what the model actually controls and the construct each score can support.

The immediate oracle exhaustively enumerates legal sequences; current-code replay completes all 200 paper-seed states below a declared 20,000-node budget (maximum 137 nodes). Combat uses a deterministic greedy reference, not optimal play. Fixed-deck fixtures contain one authored label and one on-label card among three rotated offers. The rollout is explicitly hybrid: the LLM chooses combat and rewards, while traversal and several non-combat decisions are scripted. Illegal actions and parse failures score as failures; card accuracies for unparseable outputs are not silently imputed.

Design and uncertainty. We evaluate seven configurations from five model families [6, 7, 4, 12, 11] with greedy decoding. For every available character–encoding cell, turn, combat, and fixed-deck tasks use 20 samples at each of five base seeds {42, 1042, 2042, 3042, 4042}; their sample streams do not overlap. Standard deviations are across base seeds. Hybrid runs use 20 runs per seed for the balanced small-model cohort and a separately reported 5-run floor-estimate tier for Qwen3; missing cells remain missing. Format tests use exact sign-flip permutations over matched seed pairs. With only five pairs, the minimum two-sided per-cell p is 0.0625, so inference relies on pooled strata and paired sample tests, not isolated null results.

3 Results: competence is operation-specific

Figure 1 reports native units rather than averaging non-commensurate scores. The profile immediately rejects a one-number account. DeepSeek-R1-Distill-14B has strong immediate sequencing but weak combat and a -2.73 -floor hybrid result in its one measured rollout cell. Small instruct models nearly match the greedy combat reference while differing materially in sequencing and card choice. Hybrid progress is usually within roughly one floor of seed-matched greedy; survival is at the same floor (0–3% in the balanced cohort), so we call it *on par*, not a win.

3.1 A selective Qwen3 card-choice gain

Qwen3-235B-A22B (22B active parameters, FP8 MoE) and dense Qwen3-32B share a model family but differ in architecture as well as nominal capacity; this is neither a pure capacity match nor a scaling law. Card-choice gains are $+0.256$, $+0.110$, $+0.280$, $+0.210$ for Ironclad structured, Ironclad raw, Silent structured, and Silent raw, respectively. Nineteen of 20 matched seed pairs improve. Archetype recognition rises in 15 pairs, ties in three, and falls in two. In contrast, turn differences mix signs ($+0.004$, $+0.043$, -0.021 , -0.031), combat HP differences mix signs, and hybrid floor differences are -0.64 , -0.20 , -0.12 , $+0.56$. Immediate sequencing is ceiling-limited (91–97/100 samples per cell already match the exact oracle), but that does not manufacture the card-choice result. The defensible conclusion is selective improvement in choosing a category-aligned card.

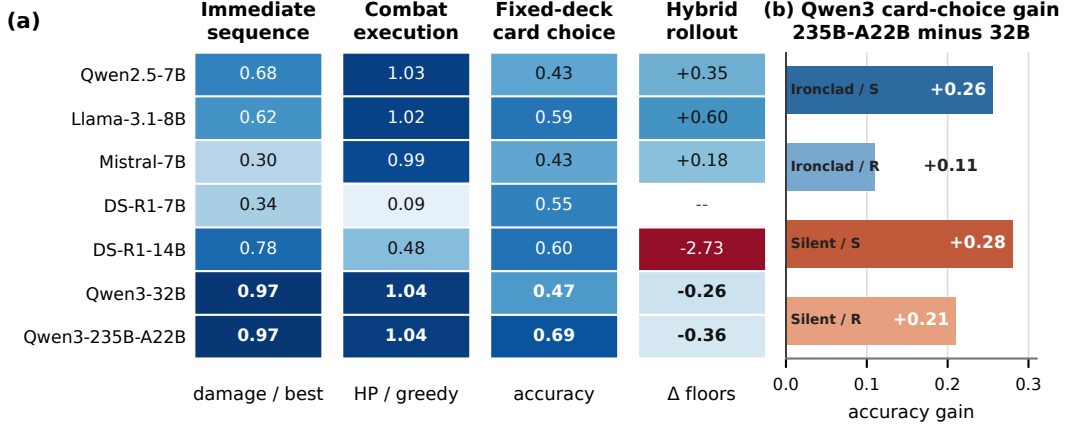


Figure 1: **Operation profiles, not a horizon curve.** (a) Each column has its own unit and scale; entries average the four character-encoding cells when available. Hybrid values are floors relative to a seed-matched greedy policy. DS-R1-14B hybrid is one cell; DS-R1-7B is unmeasured. (b) Qwen3-235B-A22B minus Qwen3-32B fixed-deck card-choice accuracy. S/R denote structured/raw. The MoE-versus-dense comparison is not a pure scaling law.

Table 2: Instrument falsification tests and their consequences.

Target	Degenerate policy	Result	Consequence
Removal-v1	always answer “Strike”	40/40 fixtures	metric quarantined
Deck label + pick	card-name dictionary; choose sole on-label offer	60/60 each; 120/120 total	bounds score as recognition
Cross-task scalar	reweight incomparable task units	weight-dependent	scalar and horizon curve retired

3.2 Encoding changes execution, not a general capability

Across 14 model-character strata, structured prompts improve combat win rate by 0.0479 (7 positive, 7 ties; pooled permutation $p = 10^{-4}$) and HP ratio by 0.0593 (12 positive, 2 negative; $p = 10^{-4}$). For immediate damage, however, the pooled magnitude favors raw by 0.0644 while directions split 6 versus 8. Fixed-deck effects also split by model. Thus representation format is part of the control interface, but its sign is not a domain-general property of “planning.”

4 Instrument falsification: what scores cannot identify

An instrument intended to measure planning should not award a perfect score to a policy that never simulates a future transition. Shortcut learning is a measurement failure when a benchmark rewards a feature that does not require the intended capability [5]. We therefore ask the cheapest adversarial question: *what would a compute-free policy score?*

Every original removal fixture has the same expert answer, Strike. More subtly, each deck contains a unique signature card and each offer contains exactly one card assigned to that label. A deterministic dictionary therefore obtains perfect archetype and pick accuracy across 20 fixtures and three rotations per character (60/60 each; 120/120 combined), without simulating a transition. These counterexamples do *not* show that the evaluated LLMs used either shortcut; they show that the instruments do not *require* planning. This necessity test, not the game-specific shortcut, is reusable: a score cannot identify a capability that an excluded control achieves perfectly. We exclude removal from every analysis and composite, rename the surviving task fixed-deck recognition/card choice, and retire the cross-task scalar and horizon plot.

Qwen3’s observed matched-seed gain remains informative under the current labels, but it supports category-aligned choice rather than forward search. Thus operation-specific differences survive while the stronger planning interpretation is falsified by the instrument itself.

97 5 A controlled protocol for measuring decision horizon

98 The falsification identifies the missing intervention. We specify a versioned same-state protocol
 99 and implement its exact-search infrastructure; model evaluation under this protocol is outside the
 100 evidence reported here. For a frozen fixture, hold the observation, legal first actions A , objective
 101 U , response contract, simulator, and first-action scoring fixed; vary only horizon $H \in \{1, 2, 4, 8\}$.
 102 Exact search assigns every legal first action an H -step value $q_H(a)$. The primary estimand is the
 103 paired within-fixture change in selected-action quality from $H = 1$ to $H = 8$, with the full slope
 104 reported. Model-blind fixtures must first pass three gates: their optimal first actions change with
 105 H often enough to create treatment strength; exact search stays within an audited budget; and an
 106 H -mismatched policy loses on the sensitive subset. This design changes the target construct without
 107 changing the task or interface.

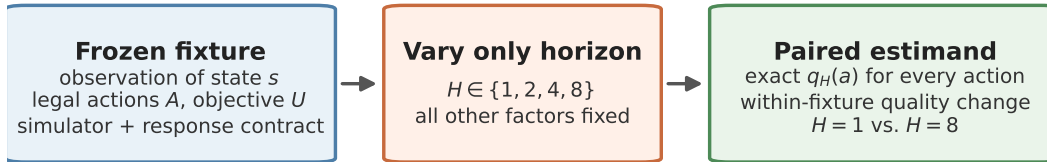


Figure 2: Controlled same-state decision-horizon protocol derived from the instrument audit. The observation, legal actions, objective, simulator, response contract, and first-action score remain fixed while only H varies. Exact search evaluates every legal first action. We specify the protocol here but claim no model-level horizon result.

108 **An audit ladder for actionable-representation benchmarks.** The completed audit suggests four
 109 gates. First, test constant, lookup, and other compute-free policies. Second, report each operation in
 110 its native unit with the model’s authority made explicit. Third, intervene on the target construct while
 111 freezing the state and interface. Fourth, validate authored utilities against downstream outcomes or
 112 an external environment. SLAY-BENCH currently clears the first two gates, specifies and implements
 113 infrastructure for the third, and does not clear the fourth. The ladder makes evidence status legible
 114 instead of hiding it in a scalar.

115 6 Limitations and conclusion

116 The simulator is deterministic but simplified and lacks conformance testing against the original game.
 117 Fixed-deck labels and utilities are authored rather than validated by downstream outcomes. Historical
 118 artifacts predate full provenance and run traces; DeepSeek-7B fixed-deck accuracies are conditioned
 119 on parseable outputs. The Qwen3 run tier has only 25 runs per cell, and the central Qwen3 comparison
 120 changes architecture as well as capacity. Seven open configurations, two characters, and one game do
 121 not establish general scaling or transfer. Finally, hybrid rollout does not measure a fully autonomous
 122 agent. The model-level controlled- H intervention is not evidence in this paper, and downstream or
 123 external validation remains incomplete.

124 Within those limits, pretrained knowledge does not become actionable uniformly, and apparently
 125 coherent benchmark profiles can arise from properties of the instrument rather than forward search.
 126 Agent benchmarks should therefore be attacked with cheap degenerate policies before their scores are
 127 interpreted as evidence of planning. Separating decision operations, reporting native units and model
 128 authority, and intervening on the intended construct provide a more defensible path from benchmark
 129 performance to capability claims.

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